1. Write and Execute code to display selected hobbies using CheckBox.

XML Code:

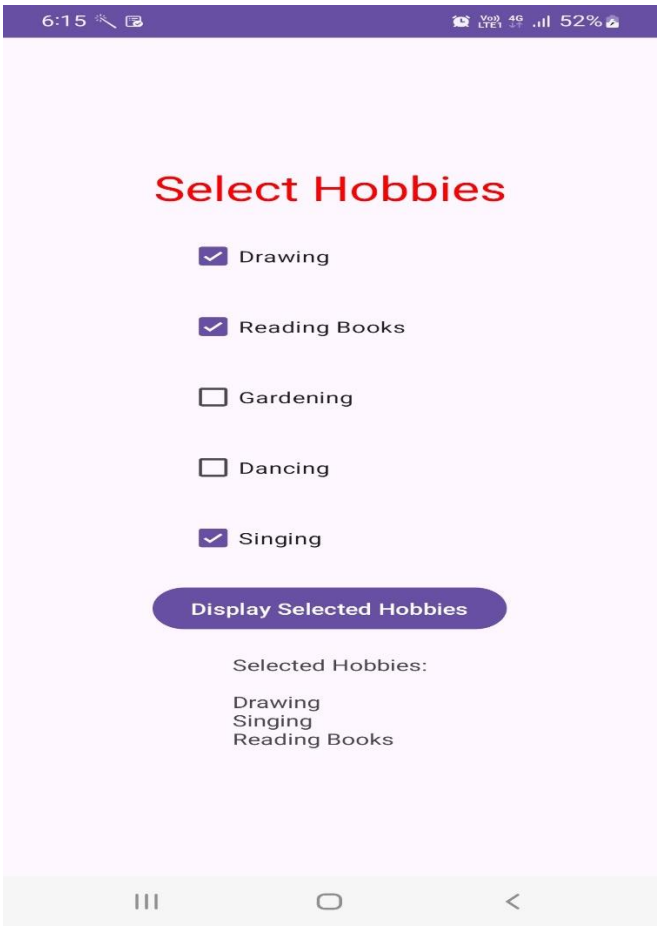
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:textSize="30sp"
        android:textColor="#f00"
        android:gravity="center_horizontal"
        android:text="Select Hobbies" />
    <CheckBox
        android:id="@+id/cnkBoxDrawing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Drawing" />

    <CheckBox
        android:id="@+id/cckBoxReadBook"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Reading Books" />

    <CheckBox
        android:id="@+id/chkBoxGarden"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Gardening" />

    <CheckBox
        android:id="@+id/chkBoxDancing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Dancing" />

    <CheckBox
        android:id="@+id/chkBoxSinging"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Singing" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Display Selected Hobbies" />

<TextView
    android:id="@+id/txtMsg"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
/>
</LinearLayout>
```

Java Code:

```
package com.example.appdemocheckboxradiobtn;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.CheckBox;
import android.widget.Toast ;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        CheckBox chkDrg = findViewById(R.id.cnkBoxDrawing );
        CheckBox chkSinging = findViewById(R.id.chkBoxSinging );
        CheckBox chkReading = findViewById(R.id.cckBoxReadBook );
        CheckBox chkGardening = findViewById(R.id.chkBoxGarden );
        CheckBox chkDance = findViewById(R.id.chkBoxDancing );

        TextView txtMsg = findViewById(R.id.txtMsg);

        Button btn = findViewById(R.id.button );

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String strMsg = "Selected Hobbies:\n";

                if (chkDrg.isChecked()) {
                    strMsg += "\n" + chkDrg.getText();
                }

                if (chkSinging.isChecked()) {
                    strMsg += "\n" + chkSinging.getText();
```

```
            }

            if (chkReading.isChecked()) {
                strMsg += "\n" + chkReading.getText();
            }
            if (chkGardening.isChecked()) {
                strMsg += "\n" + chkGardening.getText();
            }
            if (chkDance.isChecked()) {
                strMsg += "\n" + chkDance.getText();
            }
            txtMsg.setText(strMsg);
        }
    });
  }
}
```

OUTPUT:



2. Write and Execute code to count how many CheckBoxes are selected and display the count.

XML CODE:

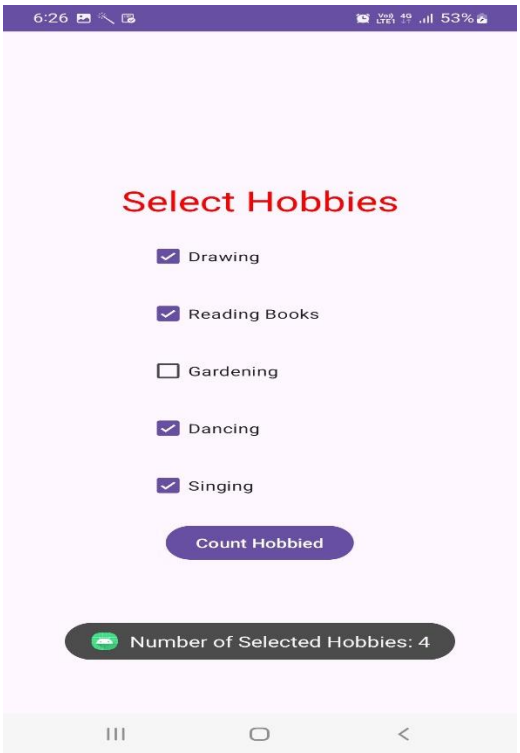
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:textSize="30sp"
        android:textColor="#f00"
        android:gravity="center_horizontal"
        android:text="Select Hobbies" />
    <CheckBox
        android:id="@+id/cnkBoxDrawing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Drawing" />
    <CheckBox
        android:id="@+id/cckBoxReadBook"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Reading Books" />

    <CheckBox
        android:id="@+id/chkBoxGarden"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Gardening" />

    <CheckBox
        android:id="@+id/chkBoxDancing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Dancing" />

    <CheckBox
        android:id="@+id/chkBoxSinging"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Singing" />

<Button
    android:id="@+id/btnCount"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Count Hobbied" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.appdemocheckboxradiobtn;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.CheckBox;
import android.widget.Toast ;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        CheckBox chkDrg = findViewById(R.id.cnkBoxDrawing );
        CheckBox chkSinging = findViewById(R.id.chkBoxSinging );
        CheckBox chkReading = findViewById(R.id.cckBoxReadBook );
        CheckBox chkGardening = findViewById(R.id.chkBoxGarden );
        CheckBox chkDance = findViewById(R.id.chkBoxDancing );

        Button btnCnt = findViewById(R.id.btnCount  );
        btnCnt.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int nCount = 0;
                if (chkDrg.isChecked()) {
                    nCount ++;
```

```
            }

            if (chkSinging.isChecked()) {
                nCount++;
            }
            if (chkReading.isChecked()) {
                nCount++;
            }
            if (chkGardening.isChecked()) {
                nCount++;
            }
            if (chkDance.isChecked()) {
                nCount++;
            }
            Toast.makeText(getApplicationContext(), "Number of Selected Hobbies: " + nCount,
Toast.LENGTH_LONG).show();
        }
    });
```

OUTPUT:




```
}
```

3. Write and Execute code to clear all CheckBox and RadioButton selections on button click.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:textSize="30sp"
        android:textColor="#f00"
        android:gravity="center_horizontal"
        android:text="Select Hobbies" />

    <CheckBox
        android:id="@+id/cnkBoxDrawing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Drawing" />

    <CheckBox
        android:id="@+id/cckBoxReadBook"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Reading Books" />

    <CheckBox
        android:id="@+id/chkBoxGarden"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"

    <CheckBox
        android:id="@+id/chkBoxSinging"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Singing" />


    <RadioButton
        android:id="@+id/radioButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:text="RadioButton" />

    <RadioButton
        android:id="@+id/radioButton2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:text="RadioButton" />

    <Button
        android:id="@+id/btnClear"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Clear All" />

</LinearLayout>
```

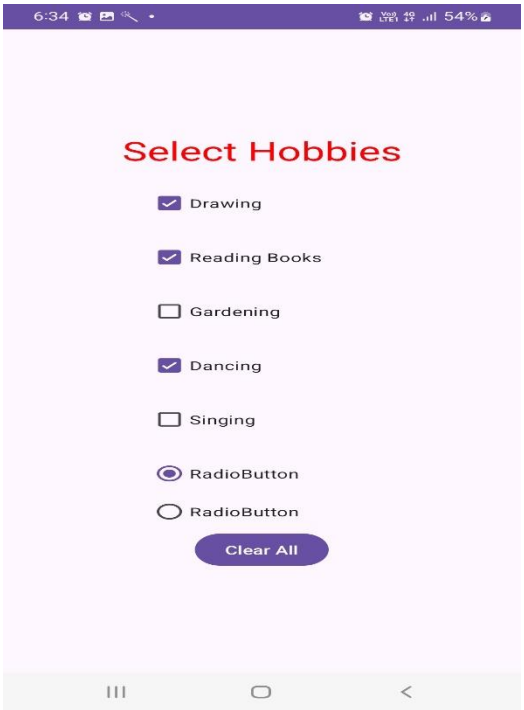
```
        android:text="Gardening" />

    <CheckBox
        android:id="@+id/chkBoxDancing"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:ems="10"
        android:text="Dancing" />
```

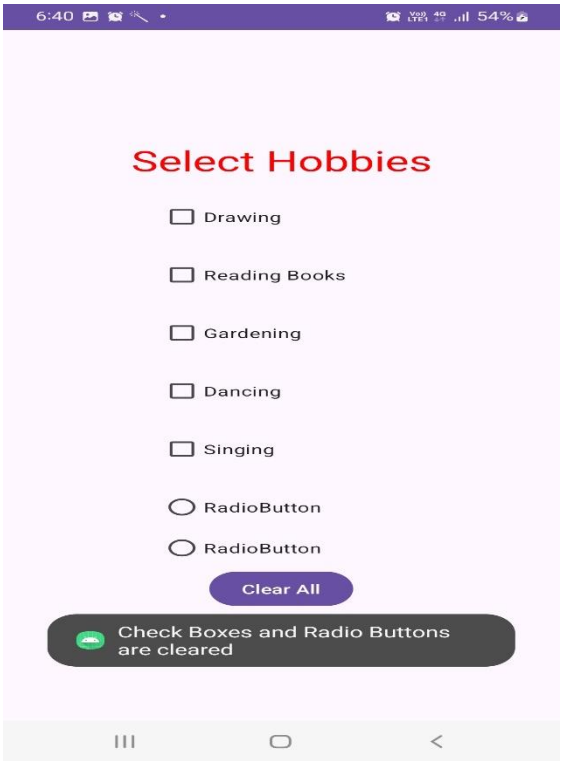
JAVA CODE:

```
package com.example.appdemocheckboxradiobtn;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.CheckBox;
import android.widget.Toast ;
import android.widget.Button;
import android.widget.RadioButton;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        CheckBox chkDrg = findViewById(R.id.cnkBoxDrawing );
        CheckBox chkSinging = findViewById(R.id.chkBoxSinging );
        CheckBox chkReading = findViewById(R.id.cckBoxReadBook );
        CheckBox chkGardening = findViewById(R.id.chkBoxGarden );
        CheckBox chkDance = findViewById(R.id.chkBoxDancing );
        Button btnClear = findViewById(R.id.btnClear );
        RadioButton radioBtn1 = findViewById(R.id.radioButton );
        RadioButton radioBtn2 = findViewById(R.id.radioButton2 );
        btnClear.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (chkDrg.isChecked()) {
                    chkDrg.setChecked(false);
                }
```

```
            if (chkSinging.isChecked()) {
                chkSinging.setChecked(false);;
            }
            if (chkReading.isChecked()) {
                chkReading.setChecked(false);
            }
            if (chkGardening.isChecked()) {
                chkGardening.setChecked(false);
            }
            if (chkDance.isChecked()) {
                chkDance.setChecked(false);
            }
            if (radioBtn1.isChecked()) {
                radioBtn1.setChecked(false);
            }
            if (radioBtn2.isChecked()) {
                radioBtn2.setChecked(false);
            }
        }
    });
  }
}
```

OUTPUT:





4. Write code to change the text color of a RadioButton when selected.

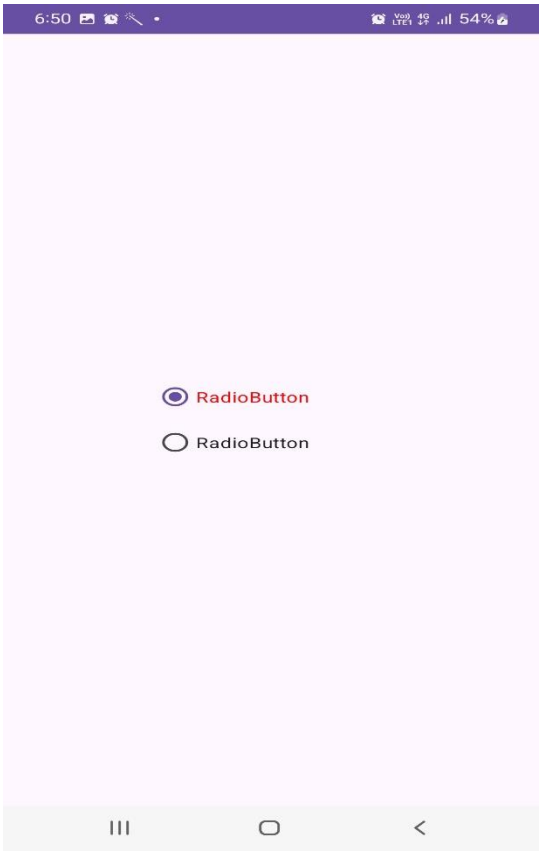
XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

<RadioGroup
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">

    <RadioButton
        android:id="@+id/radioButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:text="RadioButton" />

    <RadioButton
        android:id="@+id/radioButton2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:text="RadioButton" />

</RadioGroup>
</LinearLayout>
```

JAVA CODE:

```
package com.example.appdemocheckboxradiobtn;

import android.graphics.Color;
```

```
import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.CheckBox;
import android.widget.CompoundButton;
import android.widget.Toast ;
import android.widget.Button;
import android.widget.RadioButton;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

RadioButton radioBtn1 = findViewById(R.id.radioButton );
RadioButton radioBtn2 = findViewById(R.id.radioButton2 );

radioBtn1.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
    @Override
    public void onCheckedChanged(@NonNull CompoundButton buttonView, boolean isChecked) {
        if (isChecked ) {
            radioBtn1.setTextColor(Color.RED);
        }
        else {
            radioBtn1.setTextColor(Color.BLACK);
        }
    }
});

radioBtn2.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
    @Override
    public void onCheckedChanged(@NonNull CompoundButton buttonView, boolean isChecked) {

        if (isChecked ) {
            radioBtn2.setTextColor(Color.RED);
        }
        else {
            radioBtn2.setTextColor(Color.BLACK);
        }
```

```
    }
});
    }
}
```

OUTPUT: